

# Current Opinion in Plant Biology

## NO- and H<sub>2</sub>S-derived post-translational modifications and stress resilience

--Manuscript Draft--

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| Abstract:                           | Nitric oxide (NO) and hydrogen sulfide (H <sub>2</sub> S) have emerged as key gaseous signaling molecules in plants, playing central roles in the regulation of responses to a myriad of (a)biotic stresses. A major mechanism through which these molecules exert their effects is via post-translational modifications (PTMs) of proteins, including tyrosine nitration, S-nitrosation, and persulfidation. The last two corresponds to redox-based modifications (oxiPTMs) that can rapidly and reversibly alter protein activity, stability, localization, and interactions, thereby enabling dynamic cellular reprogramming under stress conditions. Recent advances have highlighted the intricate crosstalk between NO- and H <sub>2</sub> S-mediated PTMs, revealing a complex regulatory network that fine-tunes stress signaling pathways and enhances plant resilience. Understanding how these modifications integrate environmental cues with cellular and subcellular responses provides new insights into the molecular basis of stress tolerance and offers potential strategies for improving crop performance under adverse conditions. |
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# **NO- and H<sub>2</sub>S-derived post-translational modifications and stress resilience**

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## **Abstract**

Nitric oxide (NO) and hydrogen sulfide (H<sub>2</sub>S) have emerged as key gaseous signaling molecules in plants, playing central roles in the regulation of responses to a myriad of (a)biotic stresses. A major mechanism through which these molecules exert their effects is via post-translational modifications (PTMs) of proteins, including tyrosine nitration, S-nitrosation, and persulfidation. The last two corresponds to redox-based modifications (oxiPTMs) that can rapidly and reversibly alter protein activity, stability, localization, and interactions, thereby enabling dynamic cellular reprogramming under stress conditions. Recent advances have highlighted the intricate crosstalk between NO- and H<sub>2</sub>S-mediated PTMs, revealing a complex regulatory network that fine-tunes stress signaling pathways and enhances plant resilience. Understanding how these modifications integrate environmental cues with cellular and subcellular responses provides new insights into the molecular basis of stress tolerance and offers potential strategies for improving crop performance under adverse conditions.

**Key words:** Denitrosation, Depersulfidation, Nitration, oxi-PTMs, Persulfidation, S-nitrosation,

## Introduction

Throughout their developmental cycle, plants are exposed to a wide range of adverse abiotic factors that can negatively affect their growth and productivity. To cope with these challenges, they have developed complex regulatory networks involving diverse subcellular and cellular components that orchestrate appropriate responses to minimize potential damage [Kumar et al., 2025; Kaya and Corpas, 2026]. These responses include hormonal signaling, fluctuations in calcium levels, activation of transcription factors, and the controlled production of reactive oxygen species (ROS). Among the molecules mobilized during these stress responses, two gaseous compounds, nitric oxide (NO) and hydrogen sulfide (H<sub>2</sub>S), have emerged as key regulators over the past two decades, despite having been historically regarded as toxic [Kimura, 2026]. Both are now recognized as gasotransmitters, a class of small signaling molecules with important roles in plant physiology and in response to stresses [Corpas et al., 2026; Mohanty et al., 2026; Zhou et al., 2026]. This review discusses how NO and H<sub>2</sub>S, through their associated post-translational modifications (PTMs), influence plant responses to abiotic stress.

## Biochemical basis of PTMs mediated by NO and H<sub>2</sub>S

NO and H<sub>2</sub>S give rise to families of derived compounds known as reactive nitrogen species (RNS) and reactive sulfur species (RSS), respectively. These molecules exhibit a far more complex biochemistry than might be expected from their simple structures. One of their principal regulatory mechanisms involves interaction with specific amino acid residues, particularly tyrosine (Tyr), tryptophan (Trp), and cysteine (Cys), in target proteins, leading to post-translational modifications (PTMs) that modulate protein function. Among these, the most extensively studied in higher plants are protein Tyr-nitration and S-nitrosation, mediated by NO, and persulfidation, mediated by H<sub>2</sub>S. A key aspect of the regulatory roles of NO and H<sub>2</sub>S is their spatial specificity, as their production, metabolism, and signaling are tightly controlled through subcellular compartmentalization, allowing precise modulation of target proteins and subcellular/cellular processes.

## Protein nitration

Protein nitration involves the addition of a nitro (–NO<sub>2</sub>) group to one of the ortho carbons of the aromatic ring in tyrosine (Tyr) or tryptophan (Trp) residues. This modification typically occurs through RNS such as peroxynitrite (ONOO<sup>–</sup>) and/or the nitrogen dioxide radical (•NO<sub>2</sub>) [Bartesaghi S, Radi, 2018] (Figure 1A). In higher plants, tyrosine nitration has been

the most extensively studied, particularly at the 3-ortho position relative to the hydroxyl group at C4 of the phenolic ring. This modification can significantly alter protein function, as incorporation of the  $-\text{NO}_2$  group lowers the pKa of the phenolic group from 10.1 in Tyr to approximately 7.2 in nitrotyrosine [Sokolovsky et al., 1967; Abello et al., 2009]. Notably, nitration is a selective process: only a subset of Tyr residues within a target protein is typically modified. For example, in the pea cytosolic ascorbate peroxidase (APX), there are a total of seven Tyr residues, but only Tyr5 and Tyr235 are exclusively nitrated. Notably, the nitration of Tyr235, which is located 3.6 Å from the heme group, appears to play a more critical role in inhibiting the enzyme's activity [Begara-Morales et al., 2014]. Similarly, in pepper cytosolic L-cysteine desulfhydrase (LCD), only Tyr82 and Tyr254, out of a total of eleven Tyr residues, are nitrated by  $\text{ONOO}^-$ . These nitrated residues are situated near the enzyme's active site, and their modification results in a loss of enzymatic activity [Muñoz-Vargas et al., 2023]. In the case of Trp, nitration generally occurs at the C4, C6, or C7 positions of the indole ring. Currently, increased protein nitration is widely regarded as a marker of nitro-oxidative stress. While a specific nitrotyrosine “denitrase” activity has been described in animal systems, there is still no experimental evidence for such an activity in plants. Instead, protein nitration has been proposed to serve as a signal for protein degradation [León et al., 2022].

Beyond proteins, other biomolecules are also susceptible to nitration. These include fatty acids, such as nitro-oleic, nitro-linoleic, and nitro-linolenic acids, as well as nucleic acids (DNA, RNA, and mRNA) or  $\gamma$ -tocopherol [Mata-Pérez et al., 2016; Sytykiewicz et al., 2024; Chmielowska-Bąk et al., 2025; Desel et al., 2007]. Some of these nitrated molecules have been implicated in signaling processes and stress response, although their roles in higher plants remain poorly understood and require further investigation.

### Protein S-nitrosation

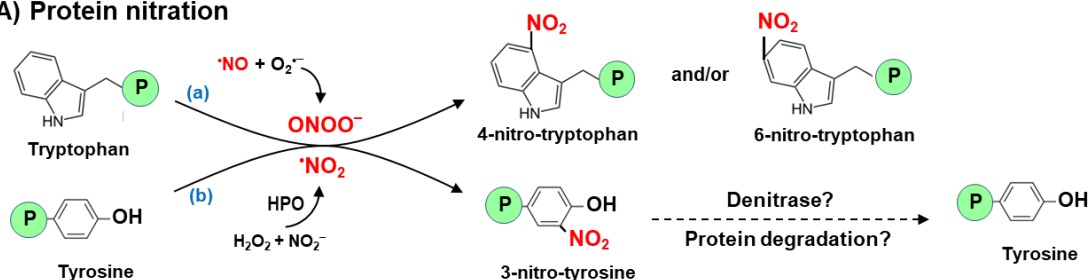
S-nitrosation, previously referred to as S-nitrosylation, involves the addition of NO to the thiol group ( $-\text{SH}$ ) of Cys residues [Gupta et al., 2020] (Figure 1B). Although proteins are the most extensively studied targets of this modification in plant cells, other molecules, such as peptides, including reduced glutathione (GSH) and free L-Cys, present at mM and  $\mu\text{M}$  concentrations, respectively, can also undergo S-nitrosation, forming S-nitrosogluthathione (GSNO) and nitrosocysteine (CysNO). These compounds play important roles in transnitrosation processes and NO-mediated signaling. In this context, S-nitrosogluthathione reductase (GSNOR), a glutathione-dependent formaldehyde dehydrogenase (FALDH), plays

a central role in regulating S-nitrosothiol levels in plant cells under physiological and stress conditions [Sakamoto et al., 2002; Lee et al., 2008; Wang et al., 2024]. Notably, GSNOR itself is subject to S-nitrosation [Guerra et al. 2016], suggesting a feedback regulatory mechanism that fine-tunes cellular NO homeostasis. Additionally, there is another mechanism of regulation of the protein S-nitrosation, which is mediated by a cytosolic thioredoxin (TRX)-mediated denitrosation and the newly identified aldo-keto reductases (AKR) [Zhang et al., 2020; Treffon et al., 2021; Treffon and Vierling, 2022; Jedelská et al., 2026].

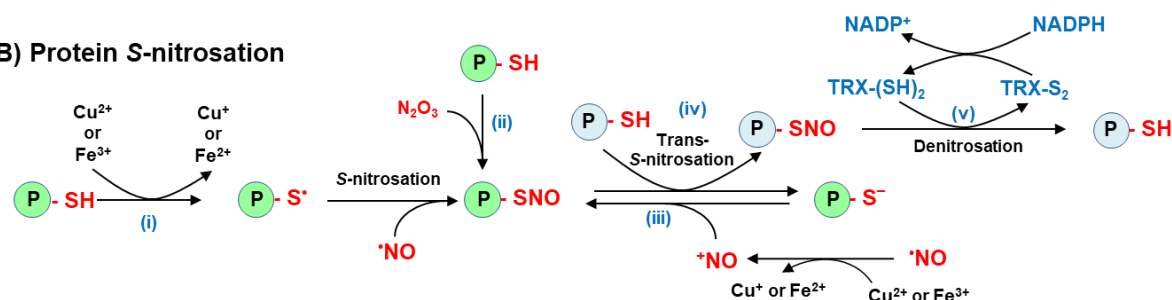
### Protein persulfidation

Persulfidation, previously named S-sulfhydration, is an H<sub>2</sub>S-derived PTM that involves the conversion of a protein Cys thiol group (–SH) into its corresponding persulfide form (–SSH) [Aroca et al., 2015; Moseler et al., 2024; Romero et al., 2025; Carrillo et al., 2026]. Persulfidation occurs not only in proteins but also leads to the formation of other molecules known as polysulfides. These include cysteine hydropersulfide (CysS–Sn–H), glutathione hydropersulfide (GS–Sn–H), and glutathione trisulfide (GSSSG) [Ogata et al., 2023]. These polysulfides are considered to be more redox-active than typical thiols and disulfides. However, research on these molecules in plants is quite limited [Kasamatsu et al., 2023a, 2023b; Ida et al., 2026]. Persulfidation can occur through several mechanisms: (i) the reaction between a protein thiol and H<sub>2</sub>S, particularly via its anionic hydrosulfide form (HS<sup>–</sup>), which is a stronger nucleophile than the protonated species; and (ii) interactions between inorganic polysulfides (–S–Sn–S–) and protein thiolates (**Figure 1C**). In addition, trans-persulfidation reactions may take place between low-molecular-weight persulfides, such as glutathione persulfide (GSSH) and cysteine persulfide (Cys–SSH), and protein persulfides, where GSH and L-Cys also contribute to this dynamic exchange network [Vignane and Filipovic 2023; Nagy et al., 2026]. Importantly, persulfidation often arises from the nucleophilic attack of a sulfide anion on oxidized forms of protein thiols, including disulfides, mixed disulfides, cysteine sulfenic acids, or S-nitrosated cysteine residues. At the cellular level, this modification is part of a broader regulatory framework in which oxiPTMs compete for the same reactive Cys residues within proteins. Under conditions of oxidative stress, modifications such as S-nitrosation and persulfidation can play a protective role by shielding these thiol groups from irreversible overoxidation to sulfinic or sulfonic acids, which is typically associated with loss of protein function [Corpas et al., 2022]. In this context, S-nitrosation and persulfidation act as reversible redox switches, preserving protein activity and contributing to the maintenance of cellular redox homeostasis.

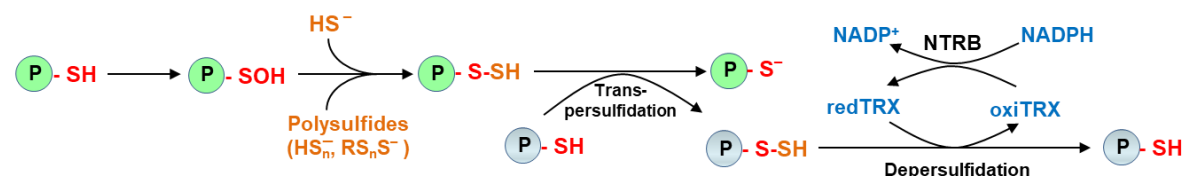
## A) Protein nitration



## B) Protein S-nitrosation



## C) Protein persulfidation

**Figure 1. NO- and H<sub>2</sub>S-derived PTMs. A) Model of protein tyrosine and tryptophan nitration.**

Two main nitrating reactions have been proposed to mediate this process: **(a)** by peroxynitrite ( $\text{ONOO}^-$ ) which is formed by the quick reaction between nitric oxide ( $\text{NO}$ ) and superoxide ( $\text{O}_2^{\cdot -}$ ) radicals; and **(b)** by nitric dioxide ( $\text{NO}_2$ ) produced through the reaction of hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) and nitrite ( $\text{NO}_2^-$ ) in the presence of a hemoperoxidase (HPO). **B) Model of protein S-nitrosation.** Thiol group ( $-\text{SH}$ ) of susceptible Cys residues of proteins can undergo S-nitrosation by different mechanisms involving different NO groups and different thiol substrates. **(i)** Thiol group can undergo a metal-catalyzed oxidation to generate a thiyl radical ( $\text{P-S}^{\cdot}$ ) with can react with  $\text{NO}$  to generate protein-SNO. **(ii)** the thiolate form of the cysteine ( $\text{PS}^-$ ) that can result from sulfur deprotonation even at physiological pH. Either as  $\text{RS}^-$ , or directly as  $\text{P-SH}$ , cysteine sulfhydryl can undergo S-nitrosation by reacting with NO-derived dinitrogen trioxide ( $\text{N}_2\text{O}_3$ ), or **(iii)** directly with nitrosonium ion ( $\text{NO}^+$ ), generated upon metal-catalyzed oxidation of  $\text{NO}$ , which acts as an electrophilic agent that triggers the S-nitrosation; **(iv)** the net transfer of  $\text{NO}^+$  from an  $\text{P-SNO}$  to an  $\text{PS}^-$  (transnitrosation) also occurs inside the cells and represents a further reaction to produce S-nitrosated adducts. **(v)** Finally, there is a mechanism of denitrosation catalyzed by a NADPH thioredoxin. **C) Model of protein persulfidation.** The cysteine thiol group ( $-\text{SH}$ ) must first be activated via oxidation to a higher oxidation state, such as sulfenic acid ( $-\text{SOH}$ ), a disulfide ( $-\text{S-S}$ ), or an S-nitrosated form ( $-\text{SNO}$ ), before it can react with  $\text{H}_2\text{S}$ . Additionally, trans-persulfidation reactions may occur with other persulfidated proteins or with low-molecular-weight persulfides, including glutathione persulfide (GSSH) and cysteine persulfide (Cys-SSH). Depersulfidation is mediated by reduced thioredoxin (red TRX), which becomes oxidized (oxTRX) during the process and is subsequently regenerated to its active reduced form by NADPH-dependent thioredoxin reductase B (NTRB) (De Brasi-Velasco et al., 2025).

## NO- and H<sub>2</sub>S-derived PTMs in abiotic stress responses

The significance of NO- and H<sub>2</sub>S-derived PTMs lies in their ability to modulate multiple levels of plant stress response mechanisms, including the regulation of antioxidant systems [Zhao et al., 2023; Wang et al., 2024; Lv et al., 2026] and an NADPH-dependent redox homeostasis [Lin et al., 2026], as well as the feedback control of NO and H<sub>2</sub>S metabolism themselves [Guerra et al., 2016; Muñoz-Vargas et al., 2023; Corpas et al., 2026]. Beyond these functions, they also participate in broader regulatory networks involving transcription factors and phytohormone signaling pathways, such as those mediated by ethylene, abscisic acid, and cytokinins, among others [Zhang et al., 2020; Chen et al., 2020; Wang et al., 2024; Wang et al., 2025; Hu et al., 2025; Sun et al., 2025; Zhang et al., 2026; Li et al., 2026]. **Table 1** summarizes representative examples of the involvement of these PTMs in plant responses to a wide range of abiotic stresses, including salinity, drought, hypoxia, senescence, cadmium toxicity, freezing, alkaline conditions, and osmotic stress. Thus, the available information support that nitration, S-nitrosation and persulfidation, play central roles in plant adaptation to abiotic stresses. S-nitrosation regulates key processes involved in salt, hypoxia, freezing, alkaline, drought, and cadmium stress responses by modulating proteins associated with RNA processing, vesicle trafficking, hormone signaling, ion homeostasis, and proline and ethylene biosynthesis. These modifications generally enhance stress tolerance by promoting osmotic adjustment, maintaining ionic balance, and activating stress-responsive gene expression, although excessive S-nitrosation under cadmium stress can aggravate cellular damage. Likewise, persulfidation contributes significantly to drought and oxidative stress tolerance by regulating ABA signaling, stomatal closure, autophagy, ROS homeostasis, and antioxidant defenses. Key protein targets include DES1, RBOHD, SnRK2.6, ABI4, and ATG4, whose persulfidation coordinates water conservation, redox balance, and stress signaling. Overall, both S-nitrosation and persulfidation act as dynamic regulatory mechanisms that integrate redox, hormonal, and metabolic pathways to enhance plant resilience under abiotic conditions.

**Table 1.** Representative protein targets of NO- and H<sub>2</sub>S-mediated post-translational modifications (nitration, S-nitrosation, and persulfidation) identified in diverse plant species under abiotic stress conditions. Abbreviations: ABA, abscisic acid; APC, active photorespiratory conditions; APX, ascorbate peroxidase; CAT, catalase; G6PDH, glucose-6-phosphate dehydrogenase; GSNO, S-nitrosoglutathione; GSNOR, S-nitrosoglutathione reductase; HRW, hydrogen-rich water; SNP, sodium nitroprusside; LCD, L-cysteine desulhydrase; MDAR, monodehydroascorbate reductase; MS, Murashige & Skoog medium; NO, nitric oxide; NPC, non-photorespiratory conditions; OE, overexpression line; ONOO<sup>-</sup>, peroxynitrite; PCD, programmed cell death; PTM, post-translational

183 modification; RBOHD, respiratory burst oxidase homologue D; RNS, reactive nitrogen species; ROS,  
1 184 reactive oxygen species; SNO, S-nitrosothiol; WT, wild type.



| Abiotic stress                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Plant species and organs                                                                                                                                                                                                               | Protein target (identified Cys)                                                | Main effects                                                                                                                                                                                                                                                                                                                                                                                          | Ref.             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>NO-derived PT Ms: Nitration / S-nitrosation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                        |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                       |                  |
| <b>Salt stress</b><br>(100, 150 or 200 mM NaCl applied on half-strength MS medium)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 to 11-day-old seedlings of <i>Arabidopsis thaliana</i> wild type (WT) and <i>prmt5-1</i> and <i>gsnor1-3</i> mutant lines.                                                                                                           | <b>Protein arginine methyltransferase 5 (PRMT5)</b><br>(Cys125)                | During salt stress, S-nitrosation induced by NO enhances the activity of the PRMT5 methyltransferase. This process generates symmetric dimethylation marks on L-Arg of spliceosome proteins, ensuring proper splicing of transcripts that respond to stress.                                                                                                                                          | Hu et al., 2017  |
| <b>Salt stress</b><br>(100–150 mM NaCl applied on half-strength MS medium). For chemical treatment, 50 $\mu$ M GSNO, 0.1 $\mu$ M phenylarsine oxide (PAO), or 15 $\mu$ M phosphatidylinositol-4-phosphate (PI4P) was sprayed onto the leaf surfaces of 5-day-old seedlings.                                                                                                                                                                                                                                                                                            | 5-day-old seedlings of <i>A. thaliana</i> WT and RABG3E knockdown mutant lines <i>rab7-1</i> and <i>rab7-2</i> .                                                                                                                       | <b>Ras-related small GTPase RABG3E</b> also known as RAB7<br>(Cys171)          | NO promotes the S-nitrosation of RAB7 at Cys171, which helps maintain the ion balance in salt-stressed plants, enhances the enzyme's GTPase activity, promoting vesicle trafficking and increasing its interaction with phosphatidylinositol-4-phosphate (PI4P). Exogenously applied PI4P increases vesicle trafficking and promotes salt tolerance depending on the S-nitrosation of RAB7 at Cys171. | Lin et al., 2023 |
| <b>Salt stress</b><br>(100 mM NaCl in Hoagland nutrient solution). In the first treatment, the nutrient solutions were supplemented with $\pm$ 10 $\mu$ M ACC (ethylene donor), 5 $\mu$ M aminovinylglycine (AVG, ethylene inhibitor) or 5 $\mu$ M 1-methylcyclopropene 1-MCP (ethylene antagonist) $\pm$ 100 $\mu$ M SNP/GSNO/ diethylamine NONOate DEA/NO (NO donors) or 100 $\mu$ M cPTIO (NO scavenger) $\pm$ 100 $\mu$ M sodium ferricyanide SF (SNP byproduct) or 100 $\mu$ M GSH/GSSG (GSNO), were only added to the nutrient solutions in the first treatment. | Four-leaf-stage (25–30 d) old hydroponic seedlings of tomato ( <i>Solanum lycopersicum</i> L. cv. Ailsa Craig (WT) and <i>GSNOR</i> overexpression ( <i>GSNOR-OE</i> ), RNAi ( <i>GSNOR-RNAi</i> ) and knockdown <i>ACOh4</i> mutants. | <b>1-aminocyclopropane-1-carboxylate oxidase homolog 4 (ACOh4)</b><br>(Cys172) | Salt stress induces NO production, activating ACOh4 transcription and leading to the S-nitrosation of ACOh4 at Cys172. This modification enhances ACOh4 enzymatic activity, increases ethylene synthesis, regulates Na <sup>+</sup> and H <sup>+</sup> efflux, and improves K <sup>+</sup> /Na <sup>+</sup> homeostasis, ultimately boosting salt tolerance.                                          | Liu et al., 2023 |

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| <p><b>Salt stress</b><br/>(150 mM NaCl in Hoagland solution <math>\pm</math> 10 <math>\mu</math>M GSNO (NO donor) or 10 <math>\mu</math>M cPTIO (NO scavenger) and <math>\pm</math> 0.1 <math>\mu</math>M brassinosteroid (BR) or 0.1 <math>\mu</math>M BR biosynthesis inhibitor, brassinazole (Brz) for 7 days.</p> | <p>30-day-old hydroponic seedlings of tomato cv. Micro-Tom (WT) and CRISPR/Cas9 mutant lines <i>slserk3a</i> and <i>slserk3b</i></p>  | <p>Somatic embryogenesis receptor kinases SERK3A and SERK3B, also known as BRI1-associated kinase 1 (BAK1)</p>                                                                                                                                          | <p>Saline stress activates NO, promoting the S-nitrosation of SERK3A/3B. This enhances its interaction with the brassinosteroid receptor BRI1, thereby activating brassinosteroid signaling and improving salt tolerance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Wei et al., 2024</p>                 |
| <p><b>Salt stress:</b> 150 mM NaCl in water or HRW (75% v/v; H<sub>2</sub> donor) <math>\pm</math> 10 <math>\mu</math>M GSNO (NO donor) or 10 <math>\mu</math>M cPTIO (NO scavenger) for 7 days.</p>                                                                                                                  | <p>Tomato cv. Micro-Tom seedlings at the three-leaf and one-heart stage (3-4 weeks after germination).</p>                            | <p><b>MAPK kinase MEK1</b> (Cys172)</p>                                                                                                                                                                                                                 | <p>H<sub>2</sub> stimulates an increase in NO levels and S-nitrosothiol (SNO) content under salt stress, which enhances the accumulation of S-nitrosated proteins. Specifically, MEK1 is induced and S-nitrosated at Cys172. This post-translational modification improves MEK1 activity, MEK1–GSNOR interaction, and enhances salinity tolerance.</p>                                                                                                                                                                                                                                                                                                                                                                 | <p>Fang et al., 2026</p>                |
| <p><b>Hypoxia</b><br/>Seeds sown on half-strength MS medium and cultured under 3% O<sub>2</sub> growth conditions.</p>                                                                                                                                                                                                | <p>7-day-old seedlings old seedlings of <i>A. thaliana</i> WT, and <i>gsnor1-3</i>, <i>nox1</i>, <i>atg2/5/7/10</i> mutant lines.</p> | <p><b>S-nitrosoglutathione reductase (GSNOR1)</b> (Cys10)</p>                                                                                                                                                                                           | <p>Hypoxia-induced S-nitrosylation of GSNOR1 at Cys10 leads to conformational changes that expose its autophagy-related 8 (ATG8)-interacting motif (AIM), making GSNOR1 a target for selective autophagy. The loss of GSNOR results in the accumulation of NO and increases the expression of hypoxia response genes (ADH1, PDC), thereby enhancing seed germination.</p>                                                                                                                                                                                                                                                                                                                                              | <p>Zhan et al., 2018</p>                |
| <p><b>Dark-induced leaf senescence</b> (DILS): rosette leaf 7 covered with aluminum foil; sampled at days 0, 3, 5 and 7</p>                                                                                                                                                                                           | <p>Rosette leaf 7 of 26-day-old plants (rosette stage) <i>A. thaliana</i></p>                                                         | <p>Tryptophan (Trp) nitrated proteins:<br/>-D-xylosidase 1 (BXL1)<br/>-Glyoxylate/succinic semialdehyde reductase 1 (GLYR1)<br/>-Dihydrolipoamide dehydrogenase 1 (LPD1)<br/>-Voltage-gated K<sup>+</sup> channel subunit <math>\beta</math> (KAB1)</p> | <p>A transient burst of ONOO<sup>-</sup> observed on day 3 of DILS did not lead to nitration of Trp proteins. However, the levels of nitroTyr proteins increased on day 3 of DILS and decreased on days 5 and 7. DILS entails a specific reprogramming of RNS formation, characterized by a temporary availability and bioactivity of ONOO<sup>-</sup> in nucleic acids and tyrosine-containing proteins. The simultaneous increase in RNA enriched with 8-nitroguanine (8-NO<sub>2</sub>-G) and the significant reduction in the pool of 6-nitroTrp-containing proteins involved in protein biosynthesis suggest that these changes contribute to the inhibition of protein biosynthesis during organ senescence.</p> | <p>Arasimowicz-Jelonek et al., 2022</p> |

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| <p><b>Freezing stress:</b><br/>(Decreasing temperature from 0°C to –6°C at a rate of 1°C/hour and holding at –6°C for 2.5 h. 3 days of recovery after stress. Pretreatment ± 100 µM SNP (NO donor).</p>                                                                                                                                                                                     | <p>Two-week-old seedlings of <i>A. thaliana</i> WT and <i>nox1</i>, <i>nia1 nia2</i>, <i>MKK5<sup>DD</sup></i>, <i>mpk3-2</i> and <i>mpk6-3</i> mutant lines.</p>                                                     | <p><b>Mitogen-activated protein kinase 6 (MPK6)</b><br/>(Cys201)</p>                                                                                                                                                                                                              | <p>SPEECHLESS SPCH is a transcription factor that controls stomatal development and freezing tolerance. MPK6-catalyzed phosphorylation of SPCH causes its degradation. NO-controlled S-nitrosation of MPK6 at Cys201 inhibits its kinase activity and stabilizes SPCH, thereby positively controlling stomatal development and freezing stress responses.</p>                                                                                                                                                                                                                                                                                                                                         | <p>Wang et al., 2025</p>   |
| <p><b>Sodic alkaline stress</b><br/>(75 mM NaHCO<sub>3</sub> in Hoagland's nutrient solution for 5 days)</p>                                                                                                                                                                                                                                                                                | <p>Fifteen-day-old hydroponic tomato (<i>Solanum lycopersicum</i> L.) seedlings plants of self-pollinated homozygous WT and GSNOR-knockdown (GSNOR-RNAi) lines.</p>                                                   | <p>110 up-regulated and 24 down-regulated S-nitrosated proteins characteristic of sodic alkaline stress conditions. Amongst them, <b>1-aminocyclopropane-1-carboxylate oxidase (ACO)</b> (Cys172) and <b>Pyrroline-5-carboxylate reductase (PYCR)</b> (Cys5) were identified.</p> | <p>Sodic alkaline stress triggers the production of GSNO, nitric oxide (NO), and S-nitrosothiols (SNOs), leading to the process of protein S-nitrosation. The enzyme GSNOR plays a crucial role in regulating the levels of protein S-nitrosation by scavenging these compounds, thereby modulating resistance to various stressors. In response to sodic alkaline stress, GSNOR down-regulates the S-nitrosation of PYCR, while ACO is S-nitrosylated under the same stress conditions. This regulation ensures that both enzymes remain active: ACO facilitates the final step in ethylene biosynthesis, which signals stress, and PYCR synthesizes L-proline to help maintain osmotic balance.</p> | <p>Gong &amp; Shi 2019</p> |
| <p><b>Osmotic stress:</b> 10% (w/v) PEG6000 in Hoagland nutrient solution (20-day-old hydroponic seedlings)<br/>Salt stress: 100 mM NaCl in Hoagland nutrient solution (20-day-old hydroponic seedlings) or 80% - 100% water retention soil with 0.1% NaCl (w/w) (30-day-old planted seedlings)<br/><b>Water deficit:</b> 50% - 80% water retention soil (30-day-old planted seedlings)</p> | <p>20/30-day-old hydroponic/planted seedlings of tomato cv. Ailsa Craig (WT) and <i>SIP5CR</i>-OE, <i>SIP5CR<sup>C5S</sup></i>-OE, and <i>SIP5CR<sup>C5W</sup></i>-OE (overexpression and site-directed) mutants.</p> | <p><b>Δ1-pyrroline-5-carboxylate reductase (P5CR)</b><br/>(Cys5)</p>                                                                                                                                                                                                              | <p>Drought and salt stress trigger NO production, which induces S-nitrosation of P5CR-Cys5. This process improves NAD(P)H affinity and enzyme activity, enhancing proline synthesis and increasing drought and salt tolerance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Liu et al., 2024</p>    |
| <p><b>Saline-alkali stress:</b> 75 mM NaHCO<sub>3</sub> in Hoagland nutrient solution (hydroponic seedlings) or natural saline-alkali soil (Na<sup>+</sup> 1.38-2.3 g kg<sup>-1</sup>, pH 7.6-8.16, EC 2.98- 4.71 mS cm<sup>-1</sup>) (planted seedlings).</p>                                                                                                                              | <p>30-day-old hydroponic or planted seedlings of tomato cv. Ailsa Craig (WT); <i>COMT</i>-, <i>GSNOR</i>-, or <i>HA2</i>-overexpression rootstocks, and</p>                                                           | <p><b>Plasma-membrane H<sup>+</sup>-ATPase 2 (HA2)</b><br/>(Cys206)</p>                                                                                                                                                                                                           | <p>Under saline-alkali conditions, NO triggers S-nitrosation of HA2 at cysteine 206 (Cys206), which impairs its interaction with 14-3-3 protein 1 (TFT1). This impairment leads to a reduction in HA activity, H<sup>+</sup> efflux, and overall saline-alkali tolerance. Additionally, NO enhances the transcription of caffeic acid O-methyltransferase (COMT), promoting</p>                                                                                                                                                                                                                                                                                                                       | <p>Wei et al., 2025</p>    |

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|                                                                                                                                                                                                                                                                                                  | nonnitrosylated <i>HA2</i> <sup>C206</sup><br>S site directed mutants                                                                                                                                                 |                                                                                                                                                                                                                                                                          | melatonin biosynthesis. This process helps to scavenge excess NO and alleviates the S-nitrosation of HA2, thereby restoring its function and improving tomato tolerance to saline-alkali conditions.                                                                                                                                                                                                                                                  |                    |
| <b>Cadmium stress</b><br>200 µM CdCl <sub>2</sub> in Hoagland nutrient solution after a 7-day pretreatment with NO and H <sub>2</sub> S donors (15 µM GSNO and 15 µM NaHS) or scavengers (100 µM cPTIO and 100 µM hypotaurine).                                                                  | 30-day-old hydroponic seedlings of tomato cv. Micro-Tom (WT) and GSNOR/LCD knockout (CRISPR/Cas9) overexpression (35S:HA-GSNOR and 35S:GFP-LCD), and site-directed (Cys was mutated to serine or tryptophan) mutants. | <b>S-nitrosogluthathione reductase (GSNOR)</b><br>(Cys47)<br><br><b>L-cysteine desulfhydrase (LCD)</b><br>(Cys225)                                                                                                                                                       | Cadmium-triggered NO burst S-nitrosates GSNOR-Cys47 and LCD-Cys225, reducing enzymatic activity, altering subcellular localization, and inhibiting the interaction between GSNOR and LCD. This increases Cd <sup>2+</sup> uptake and worsens programmed cell death under CdCl <sub>2</sub> stress.                                                                                                                                                    | Huang et al., 2026 |
| <b>Cadmium stress</b><br>1 µM CdCl <sub>2</sub> in distilled water for 5 days. To investigate Ca <sup>2+</sup> effect 0-1000 µM CaCl <sub>2</sub> or 100 µM EGTA (Ca <sup>2+</sup> scavenger) was added. To investigate NO effect 10 µM SNP (NO donor) or 200 µM cPTIO (NO scavenger) was added. | 5-day-old hydroponic cucumber ( <i>Cucumis sativus</i> ) seedlings with primary roots removed.                                                                                                                        | 117 S-nitrosated proteins identified, including pyruvate kinase 1 (PK), 6-phosphofructokinase (PFK), calcium-dependent protein kinase (CDPK), glyceraldehyde 3-phosphate dehydrogenase (GAPDH), calreticulin, fructose-bisphosphate aldolase or glutathione transferase. | Under Cd stress, the application of exogenous Ca <sup>2+</sup> increases the endogenous production of NO and the total content of S-nitrosothiols (SNO). This enhancement leads to greater S-nitrosation of various enzymes involved in glycolysis, redox processes, and calcium signaling, among others. As a result, it promotes the formation of adventitious roots and mitigates the inhibitory effects of Cd on root growth.                     | Niu et al., 2026   |
| <b>Cadmium stress</b><br>1 mM CdCl <sub>2</sub> in 1/4-strength Hoagland nutrient solution for 20 min after pretreatment with 50, 100 or 250 µM SNP.                                                                                                                                             | Three-leaf-stage seedlings of Chinese cabbage ( <i>Brassica rapa</i> ) var. glabra Regel cv. '14CR'.                                                                                                                  | 952 differential S-nitrosated proteins identified under Cd stress. Among these, heavy metal ATPase 1 (HMA1), copper chaperone for SOD (CCS) and protein arginine methyltransferase 10 (PRMT) were significantly up-regulated S-nitrosation-modified proteins.            | During Cd stress, the application of the NO donor sodium nitroprusside (SNP) enhances the activity of antioxidant enzymes. This increase helps reduce ROS and malondialdehyde (MDA) levels, thereby improving resistance to Cd. Additionally, under these conditions, proteins such as HMA1, CCS, and PRMT undergo significant S-nitrosation, suggesting that they may respond to heavy metal stress through post-translational modifications (PTMs). | Pan et al., 2026   |
| <b>H<sub>2</sub>S-derived PTM: Persulfidation</b>                                                                                                                                                                                                                                                |                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                    |

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| <p><b>Drought stress</b><br/>ABA treatment to simulate ABA accumulation in drought stress. Exposure to 10 <math>\mu</math>M ABA for 30 min <math>\pm</math> pretreatment of 10 <math>\mu</math>M diphenyleneiodonium DPI (inhibitor flavoenzymes) for 2 h.</p> | <p>Leaves and epidermal strips of <i>A. thaliana</i> ecotype Columbia (Col-0) (WT) and <i>des1</i>, <i>rbohD</i> and <i>des1rbohD</i> site-directed mutant lines.</p>                                                                                                                                                                                                                 | <p><b>L-cysteine desulfhydrase (DES1)</b> (Cys44 and Cys205)<br/><br/><b>Respiratory burst oxidase homologue (RBOHD)</b> (Cys825 and Cys890)</p>                                                                                                                                                                                                                                      | <p>Under drought stress, ABA accumulates, leading to the expression of DES1 in guard cells. DES1 can persulfide several downstream signaling proteins, including itself at Cys44 and Cys205, as well as NADPH oxidase RBOHD at Cys825 and Cys890. This process triggers the production of H<sub>2</sub>O<sub>2</sub> in guard cells, resulting in stomatal closure and increased drought tolerance. However, when ROS accumulate to high levels, they can cause the persulfide-oxidation of RBOHD and DES1 (–SSOnH), which leads to ABA desensitization. The oxidized persulfide can be reduced to a thiol group (–SH) by thioredoxin, creating a feedback mechanism that continuously prevents ABA signaling from being active in guard cells.</p> | <p>Shen et al., 2020</p>          |
| <p><b>Drought</b><br/>(Exposition to drought stress for 14 days and rewatering for 7 days after stress)</p>                                                                                                                                                    | <p>2-week-old seedlings of <i>A. thaliana</i> WT, <i>des1</i>, <i>ost1-3</i>, the ABA receptor quadruple mutant <i>pyr1;pyl1;pyl2;pyl4</i> transgenic strains and <i>nRK2.6</i><sup>C131SC137S(2CS)</sup>, <i>SnRK2.6</i><sup>S175A</sup>, <i>SnRK2.6</i><sup>S267A</sup>, <i>SnRK2.6</i><sup>S175A2CS</sup>, and <i>SnRK2.6</i><sup>S267A2CS</sup> site - directed mutant lines.</p> | <p><b>SNF1-related protein kinase SnRK2.6 also known as open stomata 1 (OST1)</b> (Cys131 and Cys137)</p>                                                                                                                                                                                                                                                                             | <p>H<sub>2</sub>S-mediated persulfidation of SnRK2.6 alters its structure, enhancing its kinase activity and promoting stomatal closure in Arabidopsis leaves, which increases drought resistance. The persulfidation of SnRK2.6 at Cys131 and Cys137 facilitates phosphorylation at Ser175, while phosphorylation at Ser267 positively regulates persulfidation at Cys137. This indicates a mutual influence between the persulfidation and phosphorylation of SnRK2.6. Both PTMs are essential for the activity of SnRK2.6 and its response to ABA, Ca<sup>2+</sup> influx, and drought stress.</p>                                                                                                                                               | <p>Chen et al., 2021</p>          |
| <p><b>Drought</b> for 10 days <math>\pm</math> pretreatment with 200 <math>\mu</math>M NaHS for 10 days before drought episode</p>                                                                                                                             | <p>15-day-old plants of <i>A. thaliana</i> WT and a line expressing green fluorescent protein (GFP)–ATG8a under the control of the <i>UBQ10</i> promoter</p>                                                                                                                                                                                                                          | <p>437 proteins were more persulfidated under drought stress and 450 proteins more persulfidated in control. Persulfidated proteins under drought stress participate in the response to oxidative stress (ascorbate peroxidase 1/4/6 APX1/4/6 or thioredoxin superfamily protein PRXQ e.g.) and hydrogen peroxide catabolic process (monodehydroascorbate reductase 1 MDAR1 e.g.)</p> | <p>Plants respond to drought by increasing available sulfide to combat drought-induced stress. H<sub>2</sub>S pretreatment before drought induces persulfidation, which enhances plant tolerance to drought by preventing H<sub>2</sub>O<sub>2</sub> accumulation and balancing redox homeostasis under stress.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Jurado-Flores et al., 2023</p> |

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| <b>Drought stress</b> ± pretreatment with 50 µM NaHS                                                                                                                                                                                                                                                                                                                                                                                                 | 25-day-old rice ( <i>Oryza sativa</i> L. cv Jsendra) plants                                                                                                                           | Multiple persulfidated drought-responsive proteins identified in the rice persulfidome | H <sub>2</sub> S enhances drought tolerance in rice by reprogramming the persulfidome of proteins involved in osmotic, antioxidant, and metabolic adjustments.                                                                                                                                                                                                                                                                                                                                                                                                     | Zhang et al., 2024           |
| <b>Osmotic stress</b><br>Rice seedlings were transferred to 1/2 Hogland nutrient solution with or without 20% PEG 6000 for 6 days.                                                                                                                                                                                                                                                                                                                   | 14-day-old rice                                                                                                                                                                       | <b>Transcription factor bZIP68</b> (Cys171)                                            | Persulfidation of Cys171 facilitates oligomerization of bZIP68 allowing transcriptional reprogramming in response to osmotic stress                                                                                                                                                                                                                                                                                                                                                                                                                                | Ma X et al., 2026            |
| <b>Osmotic stress</b><br>Arabidopsis exposed to 100 mM NaCl and drought stress,.                                                                                                                                                                                                                                                                                                                                                                     | <i>A. thaliana</i> roots were analyzed in 10-day-old seedling and leaves in 4-week-old.                                                                                               | <b>PAD3 (PHYTOALEXIN DEFICIENT 3)</b> (Cys440)                                         | PAD3 persulfidation enhances its cyanide (HCN) synthase activity, contributing to H <sub>2</sub> S–HCN homeostasis, respiratory regulation, and improved plant tolerance to osmotic stresses such as salinity and drought.                                                                                                                                                                                                                                                                                                                                         | Zhang W et al 2026           |
| <b>ABA treatment</b> to simulate ABA accumulation in abiotic stress and cause ABA-induced autophagy: Spray with 50 µM ABA for 3 and 6 h (30-day-old plants) or treatment with 50 µM ABA in MS medium for 3 and 6 h ± 200 µM NaHS for 1 h.<br>Nitrogen-starvation: growth for 4 days in nitrogen-deficient MS medium prepared by replacing nitrate salts with chloride salts.<br>Osmotic stress: growth for 4 days in MS medium with 300 mM mannitol. | One-week-old seedlings or 30-day-old plants of <i>A. thaliana</i> ecotype Columbia (Col-0) (WT) and the <i>atg4ab</i> mutant line.                                                    | <b>Cys protease ATG4</b> (Cys170)                                                      | Under basal conditions, endogenous H <sub>2</sub> S negatively regulates autophagy by maintaining ATG4 persulfidated at Cys170 and inhibiting its proteolytic activity. An increase in ABA, which occurs under stress, decreases persulfidation of ATG4, activating its protease activity and causing the C-terminal processing of ATG8, resulting in the de novo synthesis of autophagosomes.                                                                                                                                                                     | Laurea no-Marín et al., 2020 |
| <b>ABA treatment</b> to simulate ABA accumulation in abiotic stress: growth in half-strength MS medium supplemented with 20 µM ABA (seedlings); incubation with 10 µM ABA for 30 min (leaves and epidermal strips)<br>NaHS treatment: growth in half-strength MS medium supplemented                                                                                                                                                                 | One-month-old leaves and epidermal strips and five/seven-day-old seedlings of <i>A. thaliana</i> WT and <i>des1</i> , <i>abi4</i> , <i>des1abi4</i> and <i>mapkkk18</i> mutant lines. | <b>Transcription factor ABSCISIC ACID INSENSITIVE 4 (ABI4)</b> (Cys250)                | Stress-induced ABA activates the expression of DES1, which facilitates the persulfidation of ABI4 at Cys250. This PTM enhances ABI4's ability to bind to its specific motif in the promoter of <i>Mitogen-Activated Protein Kinase Kinase Kinase 18 (MAPKKK18)</i> , leading to increased transcription of this gene. Additionally, ABI4 can bind to the promoter of DES1, creating a regulatory loop. This process amplifies the signals derived from the MAPK cascade in response to ABA, thereby enhancing the transduction of environmental signals related to | Zhou et al., 2021            |

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| with 200 µM NaHS (seedlings);<br>incubation with 100 µM NaHS for 30<br>min (leaves and epidermal strips)                                                           |                                                                           |                                                                                                                                  | growth, development, and stress responses.                                                                                                                                                                                                                                                                                                                                                   |                       |
| <b>Oxidative stress</b> induced by CuO<br>nanoparticles (NPs): 20 or 100 mg/L<br>CuO NPs in medium strength<br>Hoagland solution ± 200 µM NaHS for<br>6 or 18 days | Ten-day-old tomato<br>( <i>S. lycopersicum</i> )<br>seedlings (cv. Liger) | <b>-Catalase</b> (CAT1) (Cys234)<br><br><b>-Ascorbate peroxidase</b> (APX1)<br>(Cys168)<br><br><b>-Peroxidase</b> (POD5) (Cys61) | CuO nanoparticles (NPs) induce phytotoxicity by triggering a<br>buildup of H <sub>2</sub> O <sub>2</sub> . H <sub>2</sub> S-mediated persulfidation increases the<br>activities of APX1 and POD5, while decreasing the activity of<br>CAT1. This modulation fine-tunes the H <sub>2</sub> O <sub>2</sub> detoxifying<br>network, enhancing resistance to CuO-NP-induced oxidative<br>stress. | Li et<br>al.,<br>2020 |

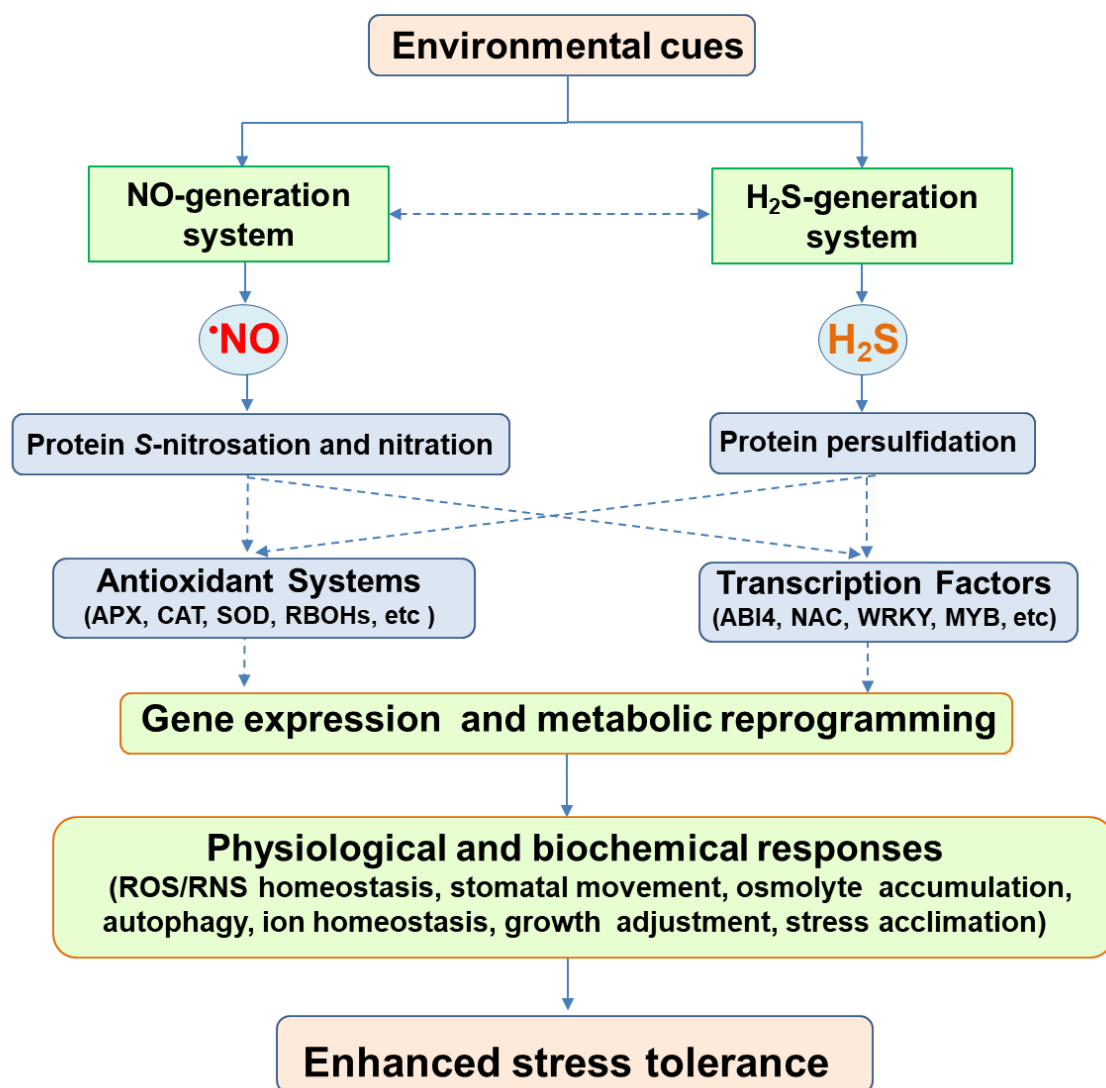
## Conclusion and future perspectives

The endogenously generated plant gas transmitters NO and H<sub>2</sub>S have been incorporated into protein signaling and regulation mechanisms in both physiological processes and in response to abiotic stresses. One of the mechanisms by which NO and H<sub>2</sub>S exert their effects is through protein PTMs, especially nitration, S-nitrosation, and persulfidation. These PTMs are specific and can either positively, negatively, or have no effect on the target protein. A particularly relevant aspect is that in many cases, a specific protein can be simultaneously and competitively regulated by several of these PTMs, either synergistically or antagonistically. The final effect on its function will influence the cellular redox state, which changes depending on the physiological status. In stress situations, there is usually an increase in ROS and RNS. Depending on the cell's capacity to respond to a specific abiotic stress, it may survive or undergo irreversible damage, leading to cell death. However, it is important to realize that plants are rarely exposed to a single abiotic stress in natural environments. Instead, they are typically subjected to multiple stresses that may occur simultaneously or sequentially over time [Jiang et al., 2025]. Consequently, plant stress responses are highly complex and rely on the integration of diverse signaling networks involving phytohormones, ROS, calcium, NO and H<sub>2</sub>S. The coordinated interplay among these signals enables plants to fine-tune their physiological and molecular responses according to the nature, intensity, and duration of the stress, as well as the developmental stage of the plant and the specific organ affected. **Figure 2** depicts a working model illustrating the signaling cascade triggered by environmental cues, in which NO and H<sub>2</sub>S act as key signaling molecules coordinating plant stress responses.

Future research should not only seek out new target proteins but also how these are regulated in a cellular context in response to abiotic stresses. Thioredoxin (TRX)-mediated denitrosation and depersulfidation are important processes that deserve further investigation. Identifying the specific TRXs involved at subcellular compartments is essential, as these small redox proteins constitute a diverse family that regulates distinct target proteins, often through NAD(P)H-dependent mechanisms. Additionally, the application of advanced techniques such as *in situ* imaging proteomics via expansion (iPEX), which combines isotropic tissue expansion with matrix-assisted laser desorption/ionization (MALDI) mass spectrometry imaging [Wang et al., 2026], holds great promise for advancing our understanding of NO- and H<sub>2</sub>S-derived PTMs. This approach could provide valuable insights into their spatial distribution and dynamics under both physiological and, particularly, stress



conditions. Furthermore, recent experimental research has shown that melatonin plays a significant role in plant signaling mechanisms involving NO and H<sub>2</sub>S under cold stress [Guo et al., 2025]. This finding opens up new avenues for research, as it highlights that the interactions among these molecules illustrate a coordinated signaling network that enhances plant adaptation and stress tolerance [Corpas et al., 2025].



**Figure 2.** Proposed model of NO- and H<sub>2</sub>S-mediated signaling during plant responses to abiotic stress. Environmental cues such as salinity, drought, temperature extremes, hypoxia, heavy metals, and osmotic stress activate NO- and H<sub>2</sub>S -generating systems. The resulting NO and H<sub>2</sub>S, together with their derived reactive nitrogen and sulfur species (RNS and RSS), trigger post-translational modifications (PTMs), primarily S-nitrosation, tyrosine nitration, and persulfidation. These PTMs regulate antioxidant and redox-related enzymes as well as stress-responsive transcription factors, leading to extensive transcriptional and metabolic reprogramming. The coordinated regulation of antioxidant defenses, hormone signaling, ion transport, autophagy, and stress-responsive gene expression ultimately promotes cellular homeostasis and enhances plant tolerance to adverse environmental conditions.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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## Data availability

No data was used for the research described in the article.

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## Highlights

- The gasotransmitters NO and H<sub>2</sub>S have signaling functions
- Both molecules coordinate plant stress responses through dynamic redox-based PTMs
- S-nitrosation and persulfidation act in concert to fine-tune protein function and stress signaling networks
- Deciphering NO- and H<sub>2</sub>S- derived PTMs provides new opportunities to enhance crop stress tolerance